

ARIS PAPADOPOULOS

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Contents

Note 3

1	<i>Review of Maths 140H</i>	5
1.1	<i>Differentiation</i>	5
1.2	<i>Integration</i>	7
2	<i>Applications of the Integral</i>	13
2.1	<i>Area Between Two Curves</i>	13
2.2	<i>Volume</i>	15
2.2.1	<i>Cross-sections</i>	15
2.2.2	<i>Solids of Revolution I</i>	16
2.2.3	<i>Solids of Revolution II</i>	18
2.3	<i>Lengths of Curves</i>	20
2.4	<i>Surface Areas</i>	22
2.5	<i>Work</i>	22
2.6	<i>Moments and Centres of Gravity</i>	25
3	<i>Parameterised Curves</i>	29
3.1	<i>Introduction to Parameterised Curves</i>	29
3.2	<i>The Length of a Parameterised Curve</i>	30
3.3	<i>The Area of a Surface (given by a parametric curve)</i>	31
4	<i>Inverse Functions</i>	35
4.1	<i>The Logarithm</i>	35
4.2	<i>Inverse Functions</i>	36

4.3	<i>Differentiation of Inverse Functions</i>	38
4.4	<i>The Natural Exponential Function</i>	40
4.5	<i>General Logarithms and Exponents</i>	40
4.6	<i>Hyperbolic Functions</i>	40
4.7	<i>Inverse Trigonometric Functions</i>	40
5	<i>L'Hôpital's Rule</i>	41
6	<i>Techniques of Integration</i>	45
6.1	<i>Basic Integrals</i>	45
7	<i>Sequences and Series</i>	49
8	<i>Curves on the Plane</i>	53
9	<i>Conic Sections</i>	55
	<i>Index</i>	59

Note

These lecture notes are not at all original. Since this is my first time teaching this course, I have heavily based them (often quoting passages) on the textbook.¹ The main other source is (naturally) Rudin's book. That being said, any mistakes here are totally my fault.

Preparing the notes was a mental aid for me, and I hope they provide some help to you, too.

¹ **CALCULUS**
w/ Concepts in Calculus
Sixth Edition
Robert Ellis | Denny Gulick

1

Review of Maths 140H

1.1 Differentiation

One of the most important concepts in calculus is that of the *derivative* (the rate of change of a quantity blah blah... you learned all about this in Maths 140). Let's remind ourselves of the (formal) definition:

1.1.1 Definition. Let $f : [a, b] \rightarrow \mathbb{R}$ be a function. For any $x \in [a, b]$ form the quotient

$$\phi(t) := \frac{f(t) - f(x)}{x - t}$$

for $t \in [a, b] \setminus \{x\}$. The derivative of f at x is defined as:

$$\lim_{t \rightarrow x} \phi(t),$$

provided the limit exists. In this case, we say that f is differentiable at x .

1.1.2 Remark. Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is differentiable at x . The picture to have in mind is that $f'(x)$ is the *slope* of the tangent to f at x . The slope of the normal to f at x is $-\frac{1}{f'(x)}$.

Differentiable functions are particularly nice. For example:

1.1.3 Theorem. Let $f : [a, b] \rightarrow \mathbb{R}$. If f is differentiable at a point $x \in [a, b]$ then f is continuous at x .¹

¹ Recall that f is continuous at x if $\lim_{y \rightarrow x} f(y) = f(x)$.

Proof. We have that

$$\lim_{x \rightarrow t} f(t) - f(x) = \lim_{x \rightarrow t} \frac{f(t) - f(x)}{t - x} \cdot (t - x) = f'(x) \cdot 0 = 0.$$

□

The converse of the theorem above (i.e. the statement “if $f : [a, b] \rightarrow \mathbb{R}$ is continuous at a point $x \in [a, b]$ then it is differentiable at x ”) is not true.

1.1.1 Exercise. Show that the function:

$$\begin{aligned} f : \mathbb{R} &\rightarrow \mathbb{R} \\ x &\mapsto |x| \end{aligned}$$

is continuous at 0 but not differentiable at 0. [Hint. What does it mean for a limit to exist?]

In practice, the formal definition of the derivative is rather tedious (and, if we're being honest, also difficult) to use. That's why it's important to have an arsenal of *standard derivatives*.

Function	Derivative
$c \in \mathbb{R}$	0
$x^r, r \in \mathbb{Q}$	rx^{r-1}
e^x	e^x
$\ln(x)$	$\frac{1}{x}$
$\sin(x)$	$\cos(x)$
$\cos(x)$	$-\sin(x)$
$\tan(x)$	$\sec^2(x)$

Table 1.1: Some standard derivatives.

We will expand our table of standard derivatives as the course goes on.

Having our basic toolbox of standard derivatives, it's important to know how to combine them to differentiate more interesting (i.e. complicated) functions.

1.1.4 Theorem (Rules of Differentiation). Let $f, g : [a, b] \rightarrow \mathbb{R}$, $\lambda \in \mathbb{R}$ and suppose that f and g are both differentiable at $x \in [a, b]$. Then $f + \lambda g$, fg , and $\frac{f}{g}$ are differentiable at x . Moreover:

1. Linearity: $(f + \lambda g)'(x) = f'(x) + \lambda g'(x)$
2. Product Rule: $(f \cdot g)'(x) = f'(x) \cdot g(x) + f(x) \cdot g'(x)$.
3. Quotient Rule: $(\frac{f}{g})'(x) = \frac{f'(x) \cdot g(x) - f(x) \cdot g'(x)}{g^2(x)}$, provided $g(x) \neq 0$.

The other extremely important rule of differentiation is the so-called *chain rule*.

1.1.5 Theorem (The Chain Rule). Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is differentiable at some point $x \in [a, b]$. Let $I \subseteq \mathbb{R}$ be some interval which contains $\text{ran}(f)$ ² and suppose that g is differentiable at the point $f(x)$. Then, $g \circ f$ is differentiable at x . Moreover:

$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x).$$

² $\text{ran}(f)$ denotes the *range* of f , i.e. all $y \in \mathbb{R}$ for which there is some $x \in [a, b]$ such that $f(x) = y$.

1.1.2 Exercise. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined by:

$$f : x \mapsto \begin{cases} x \sin\left(\frac{1}{x}\right) & x \neq 0 \\ 0 & x = 0. \end{cases}$$

1. Using the basic rules of differentiation find $f'(x)$ when $x \neq 0$.
2. Using the formal definition decide if $f'(0)$ exists (i.e. if f is differentiable at 0).

1.1.3 Exercise. Repeat the previous exercise with the function $g : \mathbb{R} \rightarrow \mathbb{R}$ be defined by:

$$g : x \mapsto \begin{cases} x \sin\left(\frac{1}{x^2}\right) & x \neq 0 \\ 0 & x = 0. \end{cases}$$

Of course, the derivative $f'(x)$ of a differentiable function $f(x)$ is also a function, so it may happen to be differentiable. In this case, we can differentiate again to obtain $f''(x)$, the second derivative. We can repeat this process as many times as we want (can), but to facilitate notation, we will write $f^{(n)}$ for the n -th derivative of f . In this notation, $f = f^{(0)}$.

1.1.4 Exercise. Find general formulas for the n -th derivative of $\sin(x)$ and $\cos(x)$.

Before moving on to reviewing integrals, let's remind ourselves of one more important definition, that of an *antiderivative*.

1.1.6 Definition. Let $f : (a, b) \rightarrow \mathbb{R}$ be a function. An antiderivative of f is a differentiable function $F : [a, b] \rightarrow \mathbb{R}$ such that for all $x \in (a, b)$ we have that $F'(x) = f(x)$.

1.1.7 Example. An antiderivative of x^4 is $\frac{1}{5}x^5 + 135$. An antiderivative of the constant function $c \in \mathbb{R}$ is cx .

You may have noticed that I keep saying *an* antiderivative rather than *the* antiderivative. That is, of course, because antiderivatives are not unique. That being said, they all look pretty similar.

1.1.8 Theorem. Suppose that F and G are antiderivatives of the same function f . Then, there is some constant $c \in \mathbb{R}$ such that $F = G + c$.

The converse of the theorem above is obvious.

1.2 Integration

The other extremely central concept of basic (single-variable) calculus is that of the *integral* (area between a curve and the x -axis blah blah...). To get us started, let's recall what a Riemann sum is.

1.2.1 Definition. Let $[a, b]$ be a given interval. A partition P of $[a, b]$ is just a finite set of points $x_0, x_1, \dots, x_n$, where:

$$a = x_0 < x_1 < \dots < x_{n-1} < x_n = b.$$

For $i \in \{1, \dots, n\}$ we write

$$\Delta x_i := x_i - x_{i-1}.$$

Now, suppose that $f : [a, b] \rightarrow \mathbb{R}$ is a bounded³ function. Corresponding to each partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ we choose some $t_i \in [x_{i-1}, x_i]$. The Riemann sum⁴ is the following quantity:

$$\sum_{i=1}^n f(t_i) \Delta x_i.$$

1.2.2 Remark. We won't care too much about the choice of $t_i \in [x_{i-1}, x_i]$, but there are some canonical choices we can make. These include the left/right endpoints of $[x_{i-1}, x_i]$,⁵ respectively, or the supremum/infimum of $f(x)$ in the interval $[x_{i-1}, x_i]$.⁶

³ This just means that $\max\{f(t) : t \in [a, b]\}, \min\{f(t) : t \in [a, b]\} \in \mathbb{R}$.

⁴ More precisely the Riemann sum corresponding to the partition P and our choice of t_i

⁵ These are the so-called left and right Riemann sums of f with respect to the partition P

⁶ These are the so called upper and lower Darboux sums of f with respect to the partition P .

We won't worry too much about what it means for a function to be (Riemann) integrable. It's enough for our purposes to know that every continuous function is integrable. Formally:

1.2.3 Theorem. For any function f continuous on an interval $[a, b]$ there is a number I such that:

For all $\epsilon > 0$ there is some $\delta > 0$ such that if $P = \{x_0, x_1, \dots, x_n\}$ is a partition of $[a, b]$ with $\Delta x_i < \delta$ for all $i \in [n]$ ⁷, then, for any choice of $t_i \in [x_{i-1}, x_i]$ we have that:

$$\left| I - \sum_{i=1}^n f(t_i) \Delta x_i \right| < \epsilon.$$

⁷ I will write $[n]$ for the set $\{1, \dots, n\}$. Some people even forego the square brackets!

1.2.4 Remark. The number I from the theorem above is unique!

1.2.5 Definition. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous,⁸. The unique number I which the Riemann sums approach in Theorem 1.2.3 is called the definite (Riemann) integral of f from a to b . We denote it by:

⁸ If I ever misspeak and say *integrable* just think "he means continuous".

$$\int_a^b f(x) \, dx.$$

1.2.1 Exercise. Let $f(x) = \sin(x)$, and let $[-\pi, \pi]$ be partitioned by $P = \{-\pi, -\frac{\pi}{2}, 0, \frac{\pi}{4}, \frac{\pi}{2}, \pi\}$. Find the left and right Riemann sums (see Remark 1.2.2) corresponding to this partition.

What should the integral $\int_{-\pi}^{\pi} \sin(x) \, dx$ be?

You may have already seen the following notation:

$$\int_a^b f(x) \, dx = \lim_{\|P\| \rightarrow 0} \sum_{i=1}^n f(t_i) \Delta x_i.$$

This is (horrible) shorthand for saying that the integral is the limit of the Riemann sums as the lengths in the partition approach 0. I'll do my best not to use this notation too often. The following lies somewhere between a theorem and a definition. I'll err on the side of the theorem...

1.2.6 Theorem. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and non-negative and let R be the region enclosed between the graph of f and the x -axis. The area of R is $\int_a^b f(x) \, dx$.

1.2.7 Example. What if f is the constant function c ? Then $\int_a^b c \, dx = c(b - a)$. What is the area of the region between the graph of $f(x) = x$ and the x -axis, from a to b ?

1.2.2 Exercise. Use the formal definition of integrals to evaluate the area of the region between the graphs of $f_1(x) = x$, $f_2(x) = x^2$ and the x -axis, from 0 to 3. That is, use the Riemann sums definition to calculate:

$$R_1 = \int_0^3 x \, dx \text{ and } R_2 = \int_0^3 x^2 \, dx.$$

For those who tried the previous exercise, it should be obvious that working with the definition of the integral through Riemann

sums is horrendous. Thankfully, the *Fundamental Theorem of Calculus* will make our lives much much easier.

First of all, the following theorem tells us that, in a sense, integration is the inverse process to differentiation:

1.2.8 Theorem. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuous and $c \in [a, b]$. Define:

$$F(x) := \int_c^x f(t) \, dt.$$

Then, F is differentiable on $[a, b]$, and:

$$F'(x) = f(x).$$

From this, we can obtain the Fundamental Theorem of Calculus:

1.2.9 Theorem (The Fundamental Theorem of Calculus). Let $f : [a, b] \rightarrow \mathbb{R}$ be a function with antiderivative $F : [a, b] \rightarrow \mathbb{R}$. Then:

$$\int_a^b f(x) \, dx = F(b) - F(a).$$

1.2.3 Exercise. Use the Fundamental Theorem of Calculus to solve Exercise 1.2.2.

The table of basic derivatives already gives you many basic integrals. Here is a recap. Throughout the table $c \in \mathbb{R}$ is some constant.

Function	Antiderivative
$d \in \mathbb{R}$	$dx + C$
$x^r, r \in \mathbb{N} \setminus \{1\}$	$\frac{1}{r+1} x^{r+1} + c$
x^{-1}	$\ln x $
e^x	$e^x + C$
$\sin(x)$	$-\cos(x) + c$
$\cos(x)$	$\sin(x) + c$

Table 1.2: Some standard antiderivatives.

We will develop more techniques to integrate later in the course.

1.2.10 Theorem.

1. Let f be continuous on $[a, b]$. Then:

$$\int_a^b f(x) \, dx = - \int_b^a f(x) \, dx.$$

2. Let f be continuous on $[a, b]$ and $[b, c]$. Then:

$$\int_a^c f(x) \, dx = \int_a^b f(x) \, dx + \int_b^c f(x) \, dx.$$

3. Let f and g be continuous on $[a, b]$, and $\lambda \in \mathbb{R}$. Then:

$$\int_a^b f(x) + \lambda g(x) \, dx = \int_a^b f(x) \, dx + \lambda \int_a^b g(x) \, dx.$$

4. Let f, g, h be differentiable functions. Then:

$$\frac{d}{dx} \int_{g(x)}^{h(x)} f(x) \, dx = f(h(x))h'(x) - f(g(x))g'(x).$$

It is common in mathematics to opt for abstraction. Here, we achieve this by considering *indefinite integrals*.

1.2.11 Definition. Let $f : [a, b] \rightarrow \mathbb{R}$ be a function that has an antiderivative on its domain. The collection of antiderivatives of f is called the indefinite integral of f , and it is denoted by:

$$\int f(x) \, dx.$$

Usually we write:

$$\int f(x) \, dx = F(x) + c,$$

for some antiderivative F of f , where c is a constant (standing for ‘any’ constant).

1.2.12 Remark. All the rules about definite integrals (that make sense⁹) from Theorem 1.2.10 still hold for indefinite integrals.

⁹ In particular, rules 3 and 4.

The most basic integration technique is the inverse of the chain rule. This is known as *integration by substitution*.

1.2.13 Theorem (Integration by Substitution). Let f and g be differentiable functions with $g \circ f$ integrable. Then:

$$\int g'(f(x))f'(x) \, dx = g(f(x)) + c.$$

This looks rather abstract, but I promise once you do enough exercises, it will make sense.

In practice. The usual trick is to substitute u for $f(x)$ and du for $f'(x) \, dx$. Let’s illustrate this with a few worked examples.

1.2.14 Example. Consider

$$\int \frac{1}{x^2} e^{\frac{1}{x}} \, dx.$$

Let $u = \frac{1}{x} = x^{-1}$. Then $du = -\frac{1}{x^2} \, dx$. Plugging that in:

$$\int \frac{1}{x^2} e^{\frac{1}{x}} \, dx = \int -e^u \, du = -e^u + c = -e^{\frac{1}{x}} + c.$$

1.2.15 Example. Consider

$$\int \frac{\tan^2 x}{x - \tan x} \, dx.$$

Recall one of the most important trigonometric identities:¹⁰

$$\cos^2(x) + \sin^2(x) = 1$$

Dividing through by $\cos^2(x)$ we get:

$$1 + \tan^2(x) = \sec^2(x),$$

i.e.

$$\tan^2(x) = \sec^2(x) - 1.$$

¹⁰ Recall that by definition, for any $t = (x, y)$ on the unit circle we have that $x = \cos(t)$ and $y = \sin(t)$, so by Pythagoras (or whatever makes sense to you) $\sin^2(t) + \cos^2(t) = 1$.

Rewriting the original integral, we see:

$$\int \frac{\tan^2(x)}{x - \tan(x)} dx = \int \frac{\sec^2(x) - 1}{x - \tan(x)} dx.$$

So let $u = x - \tan(x)$. Then, $du = (1 - \sec^2(x)) dx$, so:

$$\int \frac{\sec^2(x) - 1}{x - \tan(x)} dx = \int \frac{-1}{u} du = -\ln u + c = -\ln(1 - \tan(x)) + c.$$

1.2.16 Example. Consider

$$\int \sqrt{1 + \sqrt{x}} dx.$$

Let $u = 1 + \sqrt{x}$. Then, $du = \frac{1}{2\sqrt{x}} dx$. We have:

$$\int \sqrt{1 + \sqrt{x}} dx = \int \frac{\sqrt{x}\sqrt{1 + \sqrt{x}}}{\sqrt{x}} dx.$$

Observe that:

$$\sqrt{x} = 1 - u \text{ and } \frac{1}{\sqrt{x}} dx = 2 du,$$

so

$$\begin{aligned} \int \sqrt{1 + \sqrt{x}} dx &= 2 \int (1 - u) \sqrt{u} du \\ &= 2 \left(\int \sqrt{u} du - \int u^{\frac{3}{2}} du \right) \\ &= \frac{4}{3} u^{\frac{3}{2}} - \frac{4}{5} u^{\frac{5}{2}} + c \\ &= (1 + \sqrt{x})^2 - \frac{4}{5} (1 + \sqrt{x})^{\frac{5}{2}} + c. \end{aligned}$$

A very important application of integration by substitution is the following formula:

$$\int \frac{f'(x)}{f(x)} dx = \ln |f(x)| + c.$$

From this, we can derive many interesting integrals. For instance:

$$\int \tan(x) dx = \int \frac{\sin(x)}{\cos(x)} dx = -\ln |\cos(x)| + c$$

1.2.4 Exercise. Write out the details of the computations missing above.

1.2.17 Theorem.

$$\int \sec(x) dx = \ln |\sec(x) + \tan(x)| + c.$$

Proof. Recall¹¹ that:

$$\frac{d}{dx} \tan(x) = \sec^2(x), \text{ and } \frac{d}{dx} \sec(x) = \sec(x) \tan(x)$$

In particular:

$$\frac{d}{dx} (\sec(x) + \tan(x)) = (\sec(x))(\sec(x) + \tan(x)) \text{ Why?}$$

Now, we do something a bit silly:

$$\int \sec(x) dx = \int \frac{\sec(x)(\sec(x) + \tan(x))}{\sec(x) + \tan(x)} dx.$$

The rest of the proof is an exercise. □

¹¹ Or, even better, show these using the quotient and chain rule.

1.2.5 Exercise. Fill in the rest of the details in the proof of the previous theorem.

1.2.6 Exercise. Compute the following integrals:

1. $\int x \sin(x^2) \, dx$
2. $\int \cos(x) \sin(x) \, dx$. Do this with both substitutions $u = \cos(x)$ and $u = \sin(x)$. Compare the two answers.
3. $\int \cos(2x) \, dx$
4. $\int \sec^2(2x) \, dx$
5. $\int 2 \sin(2x)(1 - \cos(2x)) \, dx$
6. $\int \frac{\ln x}{x} \, dx$
7. $\int \frac{\ln(\ln(x))}{x \ln(x)} \, dx$
8. $\int \frac{1}{1+x^{\frac{1}{3}}} \, dx$
9. $\int \frac{x}{1+x \tan x} \, dx$
10. $\int dx$

2

Applications of the Integral

Our first new topic for the year will be on applications of the integral (**Chapter 6** of the textbook). We will focus on three main applications:

1. Volumes (e.g. we will use integrals to formally calculate the volume of a sphere, cool stuff!).
2. Surface areas (e.g. we will use integrals to show that there is a shape of finite volume with infinite surface area, even cooler stuff!).
3. Lengths of curves (we will use integrals to formally find how long complicated paths are, rather useful).

The common theme is that all these quantities can be somehow expressed (or more precisely approximated) in terms of sums that look like Riemann sums of certain functions, and thus, really, they are just integrals.

2.1 *Area Between Two Curves*

Before we start investigating more complicated topics like volumes, we should remember something a little more basic, particularly how to find the area of a region bounded between two curves.

2.1.1 Theorem. *Let $f, g : [a, b] \rightarrow \mathbb{R}$ be integrable on $[a, b]$, with $f(x) \geq g(x)$ for all $x \in [a, b]$. The area of the region R between the graphs of f and g on $[a, b]$ is given by:*

$$\int_a^b [f(x) - g(x)] \, dx.$$

Proof. By choosing an appropriate constant c , we can assume that for all $x \in [a, b]$ we have that:

$$0 \leq g(x) + c \leq f(x) + c.$$

Then, the area between $f(x)$ and $g(x)$ is equal to the area underneath $f(x) + c$, taking away the area underneath $g(x) + c$. Thus, the

desired area is:

$$\begin{aligned}\int_a^b f(x) + c \, dx - \int_a^b g(x) + c \, dx &= \int_a^b [f(x) - g(x)] + c - c \, dx \\ &= \int_a^b [f(x) - g(x)] \, dx.\end{aligned}$$

□

This is all good and well, but what if the graphs of f and g cross more than once on $[a, b]$? Then, the region whose area we want to compute is made up of several smaller regions, each of which can be computed using the previous theorem. To compute the total area, we need to find the points of intersection of the two graphs and switch the “top” and “bottom” each time, as appropriate.

2.1.2 Example ([?, Example 3, p.355]). Let $f(x) = \sin(x)$ and $g(x) = \cos(x)$. To compute the area A between the graphs of f and g on $[0, 2\pi]$ we need to find all the points where $\sin(x) = \cos(x)$ in $[0, 2\pi]$. Dividing through by $\cos(x)$ we see that we need to solve the equation

$$\tan(x) = 1.$$

The solutions are $\frac{\pi}{4}$ and $\frac{5\pi}{4}$. In the end, we have:

$$A = \int_0^{\frac{\pi}{4}} \cos(x) - \sin(x) \, dx + \int_{\frac{\pi}{4}}^{\frac{5\pi}{4}} \sin(x) + \cos(x) \, dx + \int_{\frac{5\pi}{4}}^{2\pi} \cos(x) - \sin(x) \, dx.$$

2.1.1 Exercise. Find a general formula (in terms of n) for the area between the graphs of $\sin(x)$ and $\cos(x)$ on $[\frac{\pi}{4}, \frac{\pi}{4} + n\pi]$, for $n \in \mathbb{N}$.

[Hint. What quantity repeats?]

Recall that

$$|x| = \begin{cases} -x & \text{if } x < 0 \\ x & \text{if } 0 \leq x. \end{cases}$$

In particular, given two functions $f : [a, b] \rightarrow \mathbb{R}$ we have that:

$$|f(x) - g(x)| = \begin{cases} f(x) - g(x) & \text{if } g(x) < f(x) \\ g(x) - f(x) & \text{if } f(x) \leq g(x). \end{cases}$$

Thus, for a concise formula of the area between two curves, we have:

$$\int_a^b |f(x) - g(x)| \, dx.$$

This won't help us much in computing areas, but it looks nicer.

2.1.2 Exercise. The graphs of f and g below enclose a region.¹ Find its area.

¹ This means that the endpoints of the integrals will be the intersection points of the two functions.

1. $f(x) = x^2$ and $g(x) = x^3$.
2. $f(x) = x^4 - 2$, $g(x) = 3x^8 + x^2 - 5$.
3. $f(x) = 2 - \sqrt{x}$, $g(x) = \frac{\sqrt{x+1}}{2\sqrt{x}}$.

2.2 Volume

We will consider two kinds of objects in this section:

- Objects for which we can define (in a relatively nice way) the area of their cross-sections above points on the x -axis (or y -axis).
- So called *solids of revolution*, that is, objects obtained by revolving a curve about the x -axis (or the y -axis).

2.2.1 Cross-sections

Let's start with an example.

2.2.1 Example. Suppose we have a rectangular parallelepiped in 3D space, with side lengths w, h and l . We can place our parallelepiped in 3D space so that its length starts from some point with x -coordinate a and ends at some point with x -coordinate b (i.e. $l = b - a$). Above each point in $[a, b]$, we can intersect our parallelepiped with a plane with a plane parallel to the yz -plane. The area of the intersection will be wh , so the total volume of our parallelepiped will be:

$$V = (b - a)(wl).$$

2.2.1 Exercise. To make sure you understand what happened in the example above, consider a right circular cylinder placed horizontally on the x -axis with radius r and length h . What is its volume? Recall that the area of a circle of radius r is πr^2 .

In the example and the exercise above, we considered solids in 3D space whose cross-sections had constant area. Our goal is to generalise this process to solids whose cross-sections no longer have a constant area, but the area above each point on the x -axis is given by some relatively nice function $A(x)$.

More precisely, consider a solid object D floating about in 3D space. We want to determine its volume, V . For every $x \in [a, b]$, the plane perpendicular to the x -axis intersects D in a plane region having area $A(x)$, for some function $A : [a, b] \rightarrow \mathbb{R}$. Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. The planes parallel to xz above each point in P decompose D into n slices. Let ΔV_i be the volume of the i -th slice. Then $V = \sum_{i=1}^n \Delta V_i$. The obvious thing to do is approximate each ΔV_i by a 3D shape of length $\Delta x_i = x_i - x_{i-1}$ and constant area $A(t_i)$ above all points in $[x_{i-1}, x_i]$, for some $t_i \in [x_{i-1}, x_i]$. The point is that as Δx_i becomes small, what we end up with is the "real" volume of D .

In maths:

$$V = \sum_{i=1}^n \Delta V_i \approx \sum_{i=1}^n A(t_i) \Delta x_i.$$

This is precisely a Riemann sum for A on $[a, b]$.

Thus, what we have shown² is that if $A : [a, b] \rightarrow \mathbb{R}$ is integrable, then:

$$V = \int_a^b A(x) dx.$$

² You may take this as a definition.

2.2.2 Example. Let V be a solid whose cross-section at x has area $\cos(x)$ for all $x \in [0, \frac{\pi}{2}]$. Then:

$$V = \int_0^{\frac{\pi}{2}} \cos(x) \, dx = \sin\left(\frac{\pi}{2}\right) - \sin(0) = 1.$$

2.2.3 Example. Let D be a solid whose cross-section at x is a square with side-length $\sec(x)$, for all $x \in [0, \frac{\pi}{4}]$. Then, for each $x \in [0, \frac{\pi}{4}]$ we have that the cross-section of D has area $A(x) = \sec^2(x)$, thus

$$V = \int_0^{\frac{\pi}{4}} \sec^2(x) \, dx = \tan\left(\frac{\pi}{4}\right) - \tan(0) = 1.$$

Observe that in the previous examples we always integrated with respect to x , but sometimes it is more convenient to integrate with respect to y .

This is particularly useful when the area of the cross-sections parallel to the xz -plane is easier to compute than the area of the cross-section parallel to the yz -plane. In this case, the formula becomes:

$$V = \int_a^b A(y) \, dy.$$

2.2.4 Remark. Of course, if we can compute both $A(x)$ and $A(y)$, then the result will be the same!

2.2.5 Example. Consider a pyramid which is n units tall and has a square base, m units on a side.

Since our pyramid is n units tall and starts with a base of size m^2 , we have that at point $y \in [0, n]$, the intersection of the pyramid with a plane parallel to the xz -axis passing through y will have size:

$$A(y) = \left(\frac{n(n-y)}{m}\right)^2.$$

Why? [*Hint.* Use similar triangles to find the length of a side of the square.]

Thus:

$$V = \int_0^n \left(\frac{n(n-y)}{m}\right)^2 \, dy.$$

2.2.2 Exercise. Finish the computations in the example above.

2.2.3 Exercise. Find the volume of the following solids:

1. The base of a solid is a circle with radius n , and the cross sections perpendicular to a fixed diameter of the base are equilateral triangles.
2. The base is an equilateral triangle with altitude n . The cross sections perpendicular to a given altitude of the triangle are semicircles.

2.2.2 Solids of Revolution I

2.2.6 Definition. A solid of revolution is a 3D shape whose boundary is obtained by revolving a planar shape about the x or y axis.

2.2.7 Remark. Often, when we talk about solids of revolution, we may give them as the 3D shape obtained by revolving the *graph* of a function about some axis. In this case, what we really mean is that we are revolving the region between the graph and the axis about said axis.

2.2.8 Example. A sphere is a solid of revolution obtained by revolving a semicircle about its diameter. Similarly, a frustum of a cone is a solid of revolution obtained by revolving the area under a line about the x -axis (when that line is parallel to the x -axis, we get a cylinder).

2.2.9 Theorem. Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous (bounded) function, and D the solid of revolution obtained by revolving the graph of f about the x -axis. Let V denote the volume of D . Then:

$$V = \int_a^b \pi [f(x)]^2 dx.$$

Proof. Observe that D has circular cross-sections of radius $f(x)$. Thus $A(x) = \pi [f(x)]^2$, and the result follows. $\square$

2.2.10 Example. Let S be a sphere of radius r . We know that S is a solid of revolution obtained by revolving a semicircle about its diameter. A semi-circle of radius r is given by:

$$f(x) = \sqrt{r^2 - x^2},$$

for $x \in [-r, r]$. Thus:

$$V = \int_{-r}^r \pi (\sqrt{r^2 - x^2})^2 dx = \pi \int_{-r}^r r^2 - x^2 dx = \frac{4}{3} \pi r^3.$$

2.2.4 Exercise. Find the volumes of the solids of revolution obtained by revolving the region between the graph of the given function and the x -axis about the x -axis, on the given intervals:

- $f(x) = \sec(x)$ on $[-\frac{\pi}{4}, 0]$.
- $f(x) = \sqrt{x}$ on $[0, 1]$.
- $f(x) = x^n$, $n \neq -1$ on $[0, 1]$.

Our techniques allow us to consider more complicated solids of revolution.

2.2.11 Theorem. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be integrable functions such that for all $x \in [a, b]$ we have $g(x) \leq f(x)$. Let the V be the volume of the solid generated by revolving the region between the graphs of f and g on $[a, b]$. Then:

$$V = \int_a^b \pi [(f(x))^2 - (g(x))^2] dx.$$

Proof. The plane region between the graph of f and the x -axis on $[a, b]$ is composed of the region between the graphs of f and g on $[a, b]$ and the region between the graph of g and the x -axis on $[a, b]$. The result is then an easy consequence of things we've already discussed. $\square$

2.2.12 Example. Let $f(x) = \sqrt{x+1}$ and $g(x) = \sqrt{x-1}$ on $[1, 2]$.

Then:

$$V = \int_1^2 \pi[(f(x))^2 - g(x)^2] dx = \int 4\pi.$$

2.2.3 Solids of Revolution II

So far, we've discussed how to compute the volume of a solid of revolution obtained by revolving some area about the x -axis. This method will often fail if we try to apply it to a revolution about the y -axis (the point is that the radius will be a function of x and not a function of y). Not to worry though! We will introduce a new method for this.

To get us started, let us try to see what the volume V of a cylindrical shell which is obtained by revolving a rectangle about the y -axis.

Suppose our rectangle is bounded by the x -axis, the line $y = c$ and the lines $x = a$ and $x = b$, for $a \leq b$ and c all positive.

Recall from (the solution of) Exercise 2.2.1 that the volume of a cylinder of radius r and height h is

$$\pi r^2 h.$$

Thus, the area we want which is precisely the area of the outer cylinder (the one with radius b) take away the area of the inner cylinder (the one with radius a) is:

$$V = \pi b^2 c - \pi a^2 c = \pi c(b^2 - a^2).$$

These shells will be our building blocks (akin to the circular faces in the previous section) when we try to find the area of a solid of revolution about the y -axis.

2.2.13 Theorem. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous functions with $g(x) \leq f(x)$ for all $x \in [a, b]$, and assume that $a \geq 0$. Let V be the volume of the solid region obtained by revolving about the y -axis the region R between the graphs of f and g . Then:

$$V = \int_a^b 2\pi x(f(x) - g(x)) dx.$$

Proof. Let $P = \{x_0, \dots, x_n\}$ be a partition of $[a, b]$. As before, for each $i \in [n]$ let ΔV_i be the volume of the solid obtained by revolving about the y -axis the region R_i , between the graphs of f and g on $[x_{i-1}, x_i]$. Then:

$$V = \sum_{i=1}^n \Delta V_i.$$

For each $i \in [n]$ let $t_i \in [x_{i-1}, x_i]$ be the midpoint of the subinterval, i.e.

$$t_i = \frac{x_i + x_{i-1}}{2}$$

The volume of the cylindrical shell obtained by revolving a rectangle of height $f(t_i) - g(t_i)$ and width Δx_i about the y -axis is, as we

discussed before:

$$\begin{aligned}\pi(f(t_i) - g(t_i))(x_i^2 - x_{i-1}^2) &= \pi(f(t_i) - g(t_i))(x_i - x_{i-1})(x_i + x_{i-1}) \\ &= 2\pi t_i(f(t_i) - g(t_i))(x_i - x_{i-1}) \\ &= 2\pi t_i(f(t_i) - g(t_i))\Delta x_i\end{aligned}$$

If Δx_i is small, then:

$$\Delta V_i \approx 2\pi t_i(f(t_i) - g(t_i))\Delta x_i,$$

so

$$V \approx \sum_i = 1^n 2\pi t_i(f(t_i) - g(t_i))\Delta x_i.$$

This is, of course, just a Riemann sum, so we have that:

$$V = \int_a^b 2\pi x(f(x) - g(x)) \, dx,$$

as claimed. $\square$

This method of evaluating volumes of revolution is called the *shell method*.

2.2.14 Example. Let $f(x) = mx + c$ and $d > c$. Find the volume of the shape obtained by rotating the graph of f about the y -axis below the line $y = d$.

Shell method. First, observe that the lines $y = mx + c$ and $y = d$ intersect at the point $(\frac{d-c}{m}, d)$. So, we need to compute the integral:

$$V = \int_0^{\frac{d-c}{m}} 2\pi x(d - (mx + c)) \, dx.$$

We have:

$$\begin{aligned}V &= 2\pi \left[d\frac{x^2}{2} - m\frac{x^3}{3} - c\frac{x^2}{2} \right]_0^{\frac{d-c}{m}} \\ &= \frac{\pi}{3} \left[x^2(3d - 3c + 2mx) \right]_0^{\frac{d-c}{m}} \\ &= \frac{\pi}{3} \left(\frac{(d-c)^2}{m^2} \left(3d - 3c - 2m \left(\frac{d-c}{m} \right) \right) \right) \\ &= \frac{\pi(d-c)^3}{3m^2},\end{aligned}$$

which is the volume of a cone of radius $\frac{d-c}{m}$ and height $c + d$.

Washer method. Exercise.

2.2.5 Exercise. Find the volume V of the solid generated by revolving about the x -axis the region between the graphs of the given equations:

1. $y = 5x$ and $y = x^2 + 2x + 2$
2. $y = x^3 + 2$ and $y = x^2 + 2x + 2$

2.2.6 Exercise. Let f be continuous and non-negative on $[a, \infty)$. If

$$\lim_{b \rightarrow \infty} \int_a^b \pi[f(x)]^2 \, dx$$

converges then we say that the volume of the solid obtained by revolving the graph of f about the x -axis is finite. Otherwise the volume is infinite.

For the following functions, determine if the volume of the solid obtained by revolving the graph of f about the x -axis is finite, and if it is, find it.

1. $f(x) = \frac{1}{x}$, for $x \geq 1$
2. $f(x) = \frac{1}{x^r}$, for $r \in \mathbb{Q}_{>1}$ ³ and $x \geq 1$
3. $f(x) = \frac{1}{x^r}$, for $r \in \mathbb{Q} \cap (0, 1)$ ⁴ and $x \geq 1$

³ This means that r is a rational number strictly bigger than 1.

⁴ This means that r is a rational number strictly between 0 and 1.

2.2.7 Exercise. Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous and that for all $x \in [a, b]$ we have that $f(x) \geq c$. Write down a formula for the volume V of the solid obtained by revolving about the line $y = c$ the region between the graph of f and the line $y = c$ on $[a, b]$.

Find the volume V of the solid obtained by revolving about the line $y = 1$ the region between the graph of the equation $y = e^{-2x}$ and the x -axis on the interval $[0, 1]$

2.2.8 Exercise. Suppose $f : [a, b] \rightarrow \mathbb{R}$ is continuous and assume that $c \leq a$. Write down a formula for the volume V of the solid obtained by revolving about the line $x = c$ the region between the graph of f and the x -axis.

2.2.9 Exercise (HARD). Let $f : [a, b] \rightarrow \mathbb{R}$ be a continuous function and $m, c \in \mathbb{R}$ such that for all $x \in [a, b]$ we have that $f(x) \geq mx + c$. Let R be the solid obtained by rotating the graph of f about the line $y = mx + c$. Find a general formula for R .

2.3 Lengths of Curves

Having seen how to use integrals to compute volumes, next, we will use integrals to compute *lengths* of curves. The basic example to have in mind is the length of a line segment:

2.3.1 Example. Suppose that $f : [a, b] \rightarrow \mathbb{R}$ is a linear function.⁵ The graph of f is a line segment, and the length L of the graph is just the distance between the two endpoints of the line:

$$(a, f(a)) \text{ and } (b, f(b)).$$

This is easy to compute using Pythagoras. Indeed:

$$L = \sqrt{(b-a)^2 + [f(b) - f(a)]^2}.$$

We shall now generalise this:

2.3.2 Theorem. Let $f : [a, b] \rightarrow \mathbb{R}$ be continuously differentiable. Then the length of the graph of f on $[a, b]$ is:

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx.$$

⁵ So $f = sx + c$, for some $s, c \in \mathbb{R}$ with $s \neq 0$.

Proof. We approximate the length of f using two tools: (1) Lengths of line segments and (2) Riemann sums.

Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. We approximate L using a bunch of lines with endpoints $(x_0, f(x_0)), (x_1, f(x_1)), \dots, (x_n, f(x_n))$. If the partition is fine enough, then the length of the portion of the graph of f joining $((x_{i-1}, f(x_{i-1})), (x_i, f(x_i)))$ is approximately equal to the length of the line segment joining said two points. Thus:

$$L \approx \sum_{i=1}^n \sqrt{(x_i - x_{i-1})^2 + [f(x_i) - f(x_{i-1})]^2}.$$

To express this as a Riemann sum, we want to get an expression of the form:

$$L \approx \sum_{i=1}^n H(t_i) \Delta x_i,$$

for some function H and some choice of $t_i \in [x_{i-1}, x_i]$. To this end, we make the following smart (i.e. not obvious at all) observation:

$$\frac{f(x_i) - f(x_{i-1})}{(x_i - x_{i-1})} = f'(t_i),$$

for some $t_i \in (x_i, x_{i-1})$. In particular:

$$f(x_i) - f(x_{i-1}) = f'(t_i)(x_i - x_{i-1}),$$

so

$$\begin{aligned} L &\approx \sum_{i=1}^n \sqrt{(x_i - x_{i-1})^2 + [f'(t_i)(x_i - x_{i-1})]^2} \\ &\approx \sum_{i=1}^n \sqrt{1 + [f'(t_i)]^2} (x_i - x_{i-1}) \end{aligned}$$

and this is indeed a Riemann sum. Thus:

$$L = \int_a^b \sqrt{1 + [f'(x)]^2} dx,$$

as claimed. $\square$

2.3.3 Example. Let $f(x) = \frac{2}{3}x^{\frac{3}{2}}$. Let L be the length of f from 1 to 4. Then:

$$L = \int_1^4 \sqrt{1 + [f'(x)]^2} dx.$$

First, we have that $f'(x) = x^{\frac{1}{2}}$, so:

$$L = \int_1^4 \sqrt{1 + x} dx.$$

We use the substitution $u = x + 1$, to get:

$$L = \int_2^5 \sqrt{u} du = \left[\frac{2}{3} u^{\frac{3}{2}} \right]_2^5 = \frac{2}{3} (5\sqrt{5} - 2\sqrt{2}).$$

2.4 Surface Areas – EXTRA TOPIC

2.4.1 Theorem. Let f be non-negative and continuously differentiable on $[a, b]$. The surface area of the solid obtained by revolving the graph of f about the x -axis is:

$$S = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx$$

Proof. This is similar to the previous proof. Instead of using line segments in our approximations, we now use *frustums of cones* (it's a cylindrical cone with its nose removed). Recall (or quickly learn) that the surface area of the frustum of a cone with base radii r_1 and r_2 and slant height l is:

$$S = 2\pi \left(\frac{r_1 + r_2}{2} \right) l = \pi(r_1 + r_2)l$$

These are our building blocks in the approximation, so we end up with the following:

$$\begin{aligned} S &\approx \sum_{i=1}^n \pi[f(x_{i-1}) + f(x_i)] \sqrt{(x_i - x_{i-1})^2 + [f(x_i) - f(x_{i-1})]^2} \\ &\approx \sum_{i=1}^n \pi[f(x_{i-1}) + f(x_i)] \sqrt{1 + [f'(t_i)]^2} (x_i - x_{i-1}) \\ &\approx \sum_{i=1}^n \pi 2f(t_i) \sqrt{1 + [f'(t_i)]^2} (x_i - x_{i-1}), \end{aligned}$$

for $t_i \in [x_{i-1}, x_i]$, given to us by the Mean Value Theorem. Converting this into a Riemann sum, we get:

$$S = \int_a^b 2\pi f(x) \sqrt{1 + [f'(x)]^2} dx,$$

as claimed. □

2.5 Work

Work is a concept from physics (ugh). The general rule is that:

$$\text{Work} = \text{Force} \times \text{Distance}$$

This is all good and well when the force exerted is constant, but what about when the force is variable?

Assumptions

1. The object on which the force acts moves in a straight line (i.e. along the x -axis), say from a to b .
2. At each point $x \in [a, b]$ a certain force $F(x)$ is exerted on the object.
3. The function F is continuous (integrable).

2.5.1 Theorem. Suppose an object moves from a to b under a force F , where $F : [a, b] \rightarrow \mathbb{R}$ is continuous. The work exerted by F on the object is:

$$W = \int_a^b F(x) \, dx.$$

Proof. The Riemann sum is simply

$$\sum_{i=1}^n \underbrace{F(t_i)}_{\text{Force}} \times \underbrace{\Delta x_i}_{\text{Distance}}.$$

□

2.5.2 Remark. It is implicit in the theorem above that work is positive if the force acts in the direction of motion of the object and negative if the force opposes the motion of the object

2.5.3 Example. Suppose we push a cart filled with sand for 100 meters. Our cart has a small hole on the bottom and (because of the hole), at distance x from the starting point we exert a decreasing force in Newtons given by:

$$F(x) = 60 \left(1 - \frac{x^2}{20000} \right).$$

Then, the work done by the our force is simply:

$$\int_0^{100} 60 \left(1 - \frac{x^2}{20000} \right) dx = 5000$$

2.5.4 Example (Hooke's Law). **Hooke's Law** says that the elastic force exerted by a spring which has been extended x units beyond its natural length is proportional to x . The force is negative since the string wants to return to its natural length. Thus, there is some $k > 0$ such that:

$$G(x) = -kx$$

The force $F(x)$ needed to hold the spring extended x units beyond its natural length is opposite to $G(x)$ and equal in magnitude, so:

$$F(x) = k(x).$$

Thus, the work required to stretch a spring from a units extended to b units is:

$$W = \int_a^b F(x) \, dx = \int_a^b kx \, dx = \frac{k}{2} [b^2 - a^2].$$

2.5.1 Exercise. Suppose that the work required to stretch a certain spring 1 centimetre beyond its natural length is 10^5 (unit of measurement). Find the work required to stretch it from 1 centimetre beyond its natural length to $n \geq 1$ centimetres beyond.

2.5.5 Example (Work and Kinetic Energy). Suppose that a ball of mass m is moving along the x -axis under a force $F(x)$ and with an

acceleration $\alpha(x)$. By **Newton's Second Law of Motion** the force and acceleration are related by the formula:

$$F(x) = m\alpha(x),$$

and, by definition:

$$\alpha(x) = \frac{dv}{dt}.$$

By the Chain Rule, since $v = \frac{dx}{dt}$ we have that:

$$m \frac{dv}{dt} = m \frac{dv}{dx} \frac{dx}{dt} = m \frac{dv}{dx} v.$$

In particular, we have that:

$$F(x) = mv \frac{dv}{dx}.$$

Now, consider:

$$W = \int_a^b F(x) dx = \int_a^b \left(mv \frac{dv}{dx} \right) dx = \int_{v(a)}^{v(b)} mv dv.$$

In particular:

$$W = \frac{1}{2} m[v(b)]^2 - \frac{1}{2} m[v(a)]^2.$$

We call the quantity $\frac{1}{2}mv$ the **kinetic energy** of the ball. Therefore, the work done by the force on the ball is just the change in kinetic energy.

2.5.2 Exercise. Consider a spaceship travelling away from the surface of the earth. Let:

- R denote the radius of the earth.
- G denote the universal gravitational constant.
- M denote the mass of the earth.
- m denote the mass of the spaceship.

Recall that **Newton's Law of Gravitation** says that the gravitational force F on the spaceship is:

$$F = -\frac{GMm}{r^2},$$

where r is the distance the spaceship has travelled from the earth's surface.

1. Find the work $W(b)$ done by the gravitational force as the spaceship travels from the earth's surface to a distance b from the centre of the earth, using the formula:

$$W(b) = \int_a^b F(r) dr.$$

2. Find the work $W(b)$ done by the gravitational force as the spaceship travels from the earth's surface to a distance b from the centre of the earth, using the discussion from the Kinetic energy example, assuming that the velocity of the spaceship at a distance r from the centre of the earth is $v(r)$.

3. Using the fact that the **potential energy** of the spaceship is defined to be $-GM\frac{m}{b}$, and the previous two parts prove the **Law of Conservation of Energy**:

The sum of the kinetic energy and the potential energy is constant for all values of b .

4. We say that the velocity of the spaceship is the **escape velocity**⁶ if $\lim_{b \rightarrow \infty} v(b) = 0$. Taking limits in your answer to (2) show that the escape velocity is:

$$v(R) = \sqrt{\frac{2GM}{R}}.$$

⁶ The smallest initial velocity the spaceship must have to still escape the earth's gravitational force.

2.6 Moments and Centres of Gravity

Deary me, more physics. The moment intuitively measures the tendency of mass to produce rotation. Say we have an object of mass m entirely concentrated on the point (x, y) in the plane. The moment of the object about the y -axis (M_y) is mx (the heavier the object is and/or the further away it is from the y -axis, the larger the magnitude of the moment). Similarly, the moment of the object about the x -axis, M_x is my .

The situation is similar when we have many points, we just take the sum of their moments. A collection of points is said to be at *equilibrium* about the y -axis (resp. x -axis) if $M_y = 0$ (resp. $M_x = 0$).

Let us suppose we have n objects with mass $m_1, \dots, m_n$ concentrated at points $(x_1, y_1), \dots, (x_n, y_n)$. As we said:

$$M_x = \sum_{i=1}^n m_i y_i, \text{ and } M_y = \sum_{i=1}^n m_i x_i.$$

Let:

$$m = \sum_{i=1}^n m_i.$$

We want to find a point $(\bar{x}, \bar{y})$ with the property that if we place an object of mass m at that point, then its moments about the x and y axes would be M_x and M_y respectively. Clearly:

$$\bar{x} = \frac{\sum_{i=1}^n m_i y_i}{m}, \bar{y} = \frac{\sum_{i=1}^n m_i x_i}{m}.$$

This point is called the *centre of gravity* of the collection of point masses.

We will use integrals to run the same process for mass that is distributed uniformly throughout a plane, e.g. the region between two continuous functions.

2.6.1 Theorem. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous functions with $g(x) \leq f(x)$ for all $x \in [a, b]$, and let R be the region between the graphs of f and g on $[a, b]$. Then, the moment of R about the x -axis is:

$$M_x = \int_a^b \frac{1}{2} ([f(x)]^2 - [g(x)]^2) dx,$$

and the moment of R about the y -axis is:

$$M_y = \int_a^b x[f(x) - g(x)] dx.$$

If R has positive area A , then the centre of mass of A is the point $(\mathbf{x}, \mathbf{y})$, where:

$$\mathbf{x} = \frac{M_y}{A}, \text{ and } \mathbf{y} = \frac{M_x}{A}.$$

We will slowly derive this theorem from somewhat basic observations. Let's keep track of our assumptions.

Assumptions

1. R is the region between the graphs of two continuous functions $f, g : [a, b] \rightarrow \mathbb{R}$, with $g \leq f$ on $[a, b]$.
2. The mass of any subregion of R is equal to its area.

Moments of Plane Regions about the y -Axis

2.6.2 Theorem. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous functions with $g(x) \leq f(x)$ for all $x \in [a, b]$, and let R be the region between the graphs of f and g on $[a, b]$. Then, the moment of R about the y -axis is:

$$M_y = \int_a^b x[f(x) - g(x)] dx.$$

Proof. Let $P = \{x_0, \dots, x_n\}$ be an arbitrary partition of $[a, b]$. For $i \in [n]$ let $t_i \in [x_{i-1}, x_i]$ be arbitrary, and let R_i be the portion of R lying between the vertical lines $x = x_{i-1}$ and $x = x_i$. If Δx_i is small then the area of R_i should be approximately $t_i(f(t_i) - g(t_i))$ and the moment of R_i should be approximately equal to the moment that would result if the entire area of R_i were concentrated on the line $x = t_i$. So, adding all these moments up, we get:

$$M_y \approx \sum_{i=1}^n t_i[f(t_i) - g(t_i)](x_i - x_{i-1}).$$

This is a Riemann sum, so we get:

$$M_y = \int_a^b x[f(x) - g(x)] dx,$$

as claimed. □

2.6.3 Remark. If $g(x) = 0$ then we get:

$$M_y = \int_a^b x f(x) dx,$$

the moment of f (also known as the expectation of f).

2.6.4 Example. The moment about the y axis of a rectangle R bounded by the lines $x = a$, $x = b$, $y = 0$ and $y = c$ is:

$$M_y = \int_a^b x(c - 0) dx = c(b - a) \left(\frac{a + b}{2} \right).$$

2.6.5 Remark. Notice that $c(b \vee a)$ is the area of R and $\frac{a+b}{2}$ is the distance from the centre of R to the y -axis. Notice also that this is the same moment that would result from having the entire mass concentrated at a point on the line $x = \frac{a+b}{2}$, midway between the lines $x = a$ and $x = b$.

Moments of Plane Regions about the x -axis By analogy with Example 2.6.4, if R is a rectangle, the moment M_x of R about the x -axis is:

$$M_x = \text{area of } R \times \text{distance from the centre of } R \text{ to the } x\text{-axis.}$$

This will be useful below.

2.6.6 Theorem. Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous functions with $g(x) \leq f(x)$ for all $x \in [a, b]$, and let R be the region between the graphs of f and g on $[a, b]$. Then, the moment of R about the x -axis is:

$$M_x = \int_a^b \frac{1}{2} \left([f(x)]^2 - [g(x)]^2 \right) dx.$$

Proof. We will prove the theorem when $g(x) = 0$ (see next exercise).

Let $P = \{x_0, x_1, \dots, x_n\}$ be a partition of $[a, b]$. For each $k \in [n]$ let R_k be the portion of R lying between the lines $x = x_{i-1}$ and $x = x_i$. If Δx_i is small and $t_k \in [x_{i-1}, x_i]$ then the moment of R_i about the x -axis is approximately equal to the moment about the x -axis of a rectangle with height $f(t_i)$ and base $[x_{i-1}, x_i]$. This is:

$$\frac{1}{2} [f(t_k)]^2 (x_i - x_{i-1}).$$

So, all in all:

$$M_x \approx \sum_{i=1}^n \frac{1}{2} f(t_i)^2 \Delta x_i,$$

which is a Riemann sum, and the result follows. $\square$

2.6.1 Exercise. Adapt the proof of the theorem above for an arbitrary $g(x)$ such that $g(x) \leq f(x)$ for all $x \in [a, b]$.

3

Parameterised Curves

3.1 Introduction to Parameterised Curves

So far, all curves we have looked at have been given as graphs of functions. We will now look at a more general way of defining planar curves.

For

3.1.1 Definition. Let $C \subseteq \mathbb{R}^2$ be a planar curve. We say that C is parameterised by the equations:

$$x = f(t) \text{ and } y = g(t),$$

for $t \in I$ if:

$$C = \{(f(t), g(t)) : t \in I\}.$$

In this case, we call t the parameter of the parameterisation and $x = f(t)$, $y = g(t)$ the parametric equations of C .

3.1.2 Remark. There are infinitely many parameterisations of a given curve. For instance, if C is parameterised by $x = f(t)$, $y = g(t)$ for $t \in I$, then it is also parameterised by $x = f(2t)$, $y = g(2t)$ for $t \in \frac{1}{2}I$.

Let's see how we can recover some curves we already know.

3.1.3 Theorem. Any set of parametric equations of the form:

$$x = a + bt, y = c + dt,$$

with $b, d \neq 0$ represents a straight line.

Proof. Observe that the equations we were given tell us that:

$$t = \frac{x - a}{b}, \text{ and } t = \frac{y - c}{d}.$$

In particular, we have that:

$$\frac{x - a}{b} = \frac{y - c}{d},$$

that is:

$$y = \frac{d}{b}x + \left(c - \frac{da}{b}\right),$$

which is the equation of a line. □

3.1.4 Theorem. Let $r > 0$. Then:

$$x = r \cos(t), y = r \sin(t)$$

for $t \in [0, 2\pi]$ gives us the parameterisation of a circle of radius r about the origin.

Proof. Observe that:

$$x^2 + y^2 = (r \cos(t))^2 + (r \sin(t))^2 = r^2(\cos^2(t) + \sin^2(t)) = r^2.$$

This is precisely the equation of a circle of radius r about the origin. $\square$

3.1.5 Example. Let $a, b > 0$, and consider the ellipse parameterised by $x = a \cos t$ and $y = b \sin t$ for $0 \leq t \leq 2\pi$. Find a representation of the ellipse in rectangular coordinates.

Observe that

$$x^2 = a^2 \cos^2(t) \text{ and } y^2 = b^2 \sin^2(t).$$

In particular:

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = \cos^2(t) + \sin^2(t) = 1.$$

3.1.6 Example. Consider the curve¹ given parametrically by:

$$x = \frac{3t}{1+t^3} \text{ and } y = \frac{3t^2}{1+t^3} \text{ for all } t \neq -1.$$

¹ Look up the Folium of Descartes!

Then, we have that:

$$x^3 + y^3 = \frac{27t^3(1+t^3)}{(1+t^3)^3} = \frac{27t^3}{(1+t^3)^2} = 3 \frac{3t}{1+t^3} \frac{3t^2}{1+t^3} = 3xy.$$

3.1.7 Remark. If g is any function, then the curve given parametrically by:

$$x = t \text{ and } y = g(t)$$

is the graph of g . Hence this business of parameterised curves is indeed more general!

3.2 The Length of a Parameterised Curve

3.2.1 Theorem. The length of a curve C given parametrically by continuously differentiable functions

$$x = f(t), y = g(t)$$

for $t \in [a, b]$ is:

$$L = \int_a^b \sqrt{[f'(t)]^2 + [g'(t)]^2} dt.$$

3.2.2 Remark. If C is parameterised by $x = t$ and $y = g(t)$, for $t \in [a, b]$, i.e. if C is the graph of $g(t)$, then the formula above simply becomes:

$$L = \int_a^b \sqrt{1 + [g'(t)]^2} dt,$$

which we already saw.

Proof. TODO $\square$

3.2.3 Fact. *The length of a (parameterised) curve is independent of the parameterisation.*

3.3 The Area of a Surface (given by a parametric curve)

3.3.1 Theorem. *Let C be a curve given parametrically by:*

$$x = f(t), y = g(t)$$

for $t \in [a, b]$ where f, g are differentiable on $[a, b]$ and g is non-negative on $[a, b]$.² The surface area of the shape generated when C is revolved about the x -axis is:

$$S = \int_a^b 2\pi g(t) \sqrt{[f'(t)]^2 + [g'(t)]^2} dt$$

² For simplicity, we will assume that each point of C corresponds to only one value of $t \in [a, b]$

3.3.2 Remark. Just like the case of computing the length of a parameterised curve, when C is given by:

$$x = t, \text{ and } y = g(t)$$

The formula in the theorem above simply becomes:

$$S = \int_a^b 2\pi g(t) \sqrt{1 + [g'(t)]^2} dt,$$

as we'd expect it to be!

Proof. TODO

□

Summary 1

Review the notes and ensure you have reviewed the following notions.

- Cross-sectional volume
Formula:
- Volume of revolution about the x -axis
Formula:
- Volume of revolution about the y -axis
Formula:
- Length of a curve
Formula:
- Area of a surface
Formula:
- Work
Formula:
- Moment about the x -axis
Formula:
- Moment about the y -axis
Formula:
- Centre of gravity
Formula:
- Length of a curve (parametric)
Formula:
- Area of surface (parametric)
Formula:

4

Inverse Functions

4.1 The Logarithm

This section is essentially a bit of a review from Maths 140, where hopefully you saw the formal definition of the natural logarithm for the first time.

Consider the function:

$$\begin{aligned} f : (0, \infty) &\rightarrow \mathbb{R} \\ x &\mapsto \frac{1}{x}. \end{aligned}$$

As we well know, this is a continuous function, so it is, in fact, integrable. We may define a function:

$$\begin{aligned} L : (0, \infty) &\rightarrow \mathbb{R} \\ x &\mapsto \int_1^x f(t) \, dt. \end{aligned}$$

Observe that $L(1) = 1$ and $L'(x) = \frac{1}{x}$.

4.1.1 Definition. The natural logarithm, denoted $\ln(x)$ is the function:

$$\begin{aligned} \ln : (0, \infty) &\rightarrow \mathbb{R} \\ x &\mapsto L(x). \end{aligned}$$

4.1.2 Lemma. Let $b > 0$. Then:

1. For all $r \in \mathbb{Q}$ we have that $\ln b^r = r \ln b$.
2. $\ln(\frac{1}{b}) = -\ln(b)$.

Proof. For the first item, let $F(x) = \ln(x^r)$ and $G(x) = r \ln(x)$. Clearly $F(1) = G(1) = 0$. Moreover:

$$F'(x) = \frac{d}{dx} \int_1^{x^r} \frac{1}{t} \, dt = \frac{1}{x^r} r x^{r-1} = r \frac{1}{x} = \frac{d}{dx} r \int_1^x \frac{1}{t} \, dt = G'(x).$$

Thus, the two functions have the same derivative and the same value at 1. In particular, they are equal.

The argument for the second item is similar. □

4.1.3 Theorem.

$$\lim_{x \rightarrow \infty} \ln(x) = \infty \text{ and } \lim_{x \rightarrow 0^+} \ln(x) = -\infty.$$

Proof. For the first limit, let $b > 1$ be arbitrary. Then, by the previous lemma:

$$\lim_{x \rightarrow \infty} \ln(x) = \lim_{r \rightarrow \infty} \ln(b^r) = \ln(b) \lim_{r \rightarrow \infty} r = \infty.$$

The second limit follows from the first and the previous lemma, since:

$$\lim_{x \rightarrow 0^+} \ln(x) = \lim_{\frac{1}{x} \rightarrow \infty} \ln(x) = \lim_{\frac{1}{x} \rightarrow \infty} -\ln\left(\frac{1}{x}\right) = -\infty.$$

□

4.1.1 Exercise. Show that the graph of $\ln(x)$ is concave down.

4.1.4 Remark. Since $\ln(1) = 0$ and $\lim_{x \rightarrow \infty} \ln(x) = \infty$, by the Intermediate Value Theorem, there is a (unique, since $\ln$ is monotone) real number e such that $\ln(e) = 1$. We'll come back to this very soon.

4.1.2 Exercise. Prove the following (well-known) statements, using the formal definition of the natural logarithm:

1. For all $b, c > 0$ we have that $\ln(bc) = \ln(b) + \ln(c)$.
2. For all $b, c > 0$ we have that $\ln\left(\frac{b}{c}\right) = \ln(b) - \ln(c)$.

4.2 Inverse Functions

Let us consider functions in the abstract for a moment. Recall that a function is a recipe that sends elements of a set X to a set Y such that every element of X is sent to a unique element of Y .¹ We call X the domain of f and Y the codomain of f . We call the set of elements of Y that can be obtained by applying f to some element of X the range of f .

¹ Elements of Y can be obtained by applying f to many different elements of X .

Let us now distinguish some rather special kinds of functions:

4.2.1 Definition. Let $f : X \rightarrow Y$ be a function. We say that f is injective if for all $x, y \in X$ we have that $f(x) = f(y)$ implies that $x = y$. We say that f is surjective if for all $y \in Y$ there is some $x \in X$ such that $y = f(x) = f(y)$. We say that f is bijective or a bijection if it is both injective and surjective.

4.2.2 Remark. It is not hard to see that $f : X \rightarrow Y$ is a bijection precisely when there is a function $g : Y \rightarrow X$ such that for all $x \in X$ and all $y \in Y$ we have that:

$$f(x) = y \text{ if, and only if, } g(y) = x.$$

4.2.3 Definition. Let $f : X \rightarrow Y$ be a bijection and suppose that $g : Y \rightarrow X$ is a function as in Remark 4.2.2. We call g the inverse of f , and denote it by f^{-1} .

4.2.4 Example. TODO

In practice. Once we know that a function has an inverse, we can compute a formula for the inverse by the following method:

1. Write $y = f(x)$
2. Solve for x in terms of y .
3. Change the roles of x and y .
4. The domain of f^{-1} is the range of f .

4.2.5 *Example* (Fahrenheit-to-Celsius function). Let $f = \frac{9}{5}x + 32$ for $x \geq -273.15$. Find a formula for the inverse of f and determine its range. We follow the steps:

$$y = \frac{9}{5}x + 32$$

$$x = \frac{5}{9}(y - 32).$$

Switching the roles of x and y we have that:

$$f^{-1}(x) = \frac{5}{9}(x - 32).$$

Finally to find the domain of $f^{-1}(x)$, we compute the range of $f(x)$. We have that $f(-273.15) = -458.67$, so the range of f is $[-458.67, \infty)$.

4.2.6 Theorem. *Let f be a bijection. Then:*

- $(f^{-1})^{-1} = f$.
- $f^{-1}(f(x)) = x$, for all x in the domain of f .
- $f(f^{-1}(x)) = x$ for all x in the range of f .

Proof. Exercise. □

We will, in general, forego the surjectivity condition, and identify the range and the codomain of real-valued functions. The point is that if $f : I \rightarrow \mathbb{R}$ is a continuous function, for an interval I then, by the Intermediate Value Theorem, there is an interval J such that $\text{ran}(f) = J$. Thus $f : I \rightarrow J$ is surjective.

The following should now be clear:

4.2.7 Theorem. *Every strictly monotone² function has an inverse.*

4.2.8 Corollary. *Let $f : I \rightarrow \mathbb{R}$ be a differentiable function. If $f'(x) > 0$ for all but finitely many interior points $x \in I$ (or if $f'(x) < 0$ for all but finitely many interior points $x \in I$), then f has an inverse on I .*

The converse of Theorem 4.2.7 is also true. More precisely:

4.2.9 Theorem. *Let f be continuous on an interval I if f is injective then f is either strictly increasing or strictly decreasing.*

Proof. Suppose not. Then there exist $a < b < c \in I$ such that $f(b)$ does not lie between $f(a)$ and $f(c)$. This means that either $f(a) < f(c) < f(b)$ or $f(b) < f(c) < f(a)$. In the first case, by the Intermediate Value Theorem, there is some $d \in [a, b]$ such that $f(d) = f(c)$ and in the second case, again by the Intermediate Value Theorem there is some $d \in [b, c]$ such that $f(d) = f(a)$. In either case, we have a contradiction. □

² Recall that a function f is strictly increasing (resp. strictly decreasing) if for all $x < y$ in the domain of f we have that $f(x) < f(y)$ (resp. $f(y) < f(x)$). We say that f is strictly monotone if it is either strictly increasing or strictly decreasing.

4.3 Differentiation of Inverse Functions

Before we get into the main theoretical results of this section, let's discuss how the graph of f^{-1} is related to the graph of f .

Let (x, y) be a point on the graph of f . By definition, this means that $f(x) = y$. Thus, the point (y, x) will be a point on the graph of $f^{-1}(x)$. Observe that the points (x, y) and (y, x) are symmetric with respect to the line $y = x$. Thus:

4.3.1 Fact. *The graph of f^{-1} is obtained by reflecting the graph of f through the line $y = x$.*

4.3.2 Theorem. *Let $f : I \rightarrow J$ be a continuous bijection. Then $f^{-1} : J \rightarrow I$ is continuous.*

Proof. Let c be an interior point of J . We must show that:

$$\lim_{x \rightarrow c} f^{-1}(x) = f^{-1}(c).$$

Expanding, we must show that for all $\varepsilon > 0$ there is a $\delta > 0$ such that whenever $|y - c| < \delta$ then $|f^{-1}(y) - f^{-1}(c)| < \varepsilon$.

Since f is invertible and continuous it must be either strictly increasing or strictly decreasing on I . Assume first that f is strictly increasing. Then:

$$f(f^{-1}(c) - \varepsilon) < f(f^{-1}(c)) = c < f(f^{-1}(c) + \varepsilon),$$

and let $\delta > 0$ be any number such that:

$$\delta < \min\{c - f(f^{-1}(c) - \varepsilon), f(f^{-1}(c) + \varepsilon) - c\}.$$

Since f^{-1} is also strictly increasing, we have that:

$$f^{-1}(c) - \varepsilon < f^{-1}(c - \delta) < c < f^{-1}(c + \delta) < f^{-1}(c) + \varepsilon$$

So, if $z = f(x) \in (c - \delta, c + \delta)$ then $x \in (f^{-1}(c - \delta), f^{-1}(c + \delta))$, and thus $x = f^{-1}(z) \in (f^{-1}(c) - \varepsilon, f^{-1}(c) + \varepsilon)$, and we are done.

When f is strictly decreasing the proof is the same with all the inequalities reversed. $\square$

Now that we know that inverses of continuous functions are continuous, we will delve deeper into differentiation of inverse functions of differentiable functions:

4.3.3 Theorem. *Suppose that f is invertible and continuous on some interval I . Suppose that for $a \in I$, we know that:*

- $f(a) = c$
- f is differentiable at a .
- $f'(a) \neq 0$.

Then, f^{-1} is differentiable at c , and:

$$\left(f^{-1}(c)\right)' = \frac{1}{f'(a)}.$$

Proof. We consider:

$$\lim_{y \rightarrow c} \frac{f^{-1}(y) - f^{-1}(c)}{y - c} = \lim_{y \rightarrow c} \frac{f^{-1}(y) - a}{f(f^{-1}(y)) - f(a)}.$$

Since f^{-1} is continuous at c we know that $\lim_{y \rightarrow c} f^{-1}(y) = f^{-1}(c) = a$. Of course, we know that $f^{-1}(y) \neq a$ for any $y \neq c$. So, substituting x for $f^{-1}(y)$ in the equation above we have:

$$\lim_{y \rightarrow c} \frac{f^{-1}(y) - f^{-1}(c)}{y - c} = \lim_{x \rightarrow a} \frac{x - a}{f(x) - f(a)} = \frac{1}{\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}}.$$

The latter limit exists, since f is differentiable at a and we see that:

$$(f^{-1}(c))' = \frac{1}{f'(a)},$$

as claimed. $\square$

4.3.4 Remark. The intuition behind the theorem above is the following:

- Both f and f^{-1} are increasing or both f and f^{-1} are decreasing.
- When f increases fast, then f^{-1} increases slow.
- When f increases slow, then f^{-1} increases fast.

The theorem we just proved can be summarised nicely in Leibniz notation:

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}},$$

where $\frac{dx}{dy}$ is the derivative of f^{-1} (the rate of change of x with respect to y).

4.3.5 Example. Let:

$$f(x) = \int_0^x \sqrt{1+t^4} dt.$$

Observe that $\sqrt{1+t^4} \geq 0$ for all $x \geq 0$. In particular, $\int_0^{x_1} \sqrt{1+t^4} dt < \int_0^{x_2} \sqrt{1+t^4} dt$ when $x_1 < x_2$, since the area under the graph of $\sqrt{1+t^4}$ increases. Now, let $c = f(1)$. To compute $(f^{-1})'(c)$ we use the theorem relating the derivative of f with the derivative of f^{-1} . By that theorem, we know that:

$$(f^{-1}(c))' = \frac{1}{f'(1)}$$

Of course, we know by the Fundamental Theorem of Calculus that:

$$f'(1) = (F(1) - F(0))',$$

for any antiderivative F of f . In particular, we have that:

$$f'(1) = \sqrt{2} - 1,$$

so

$$(f^{-1}(c))' = \frac{1 + \sqrt{2}}{3}.$$

4.3.1 Exercise. Can a polynomial of even degree have an inverse? What about a polynomial of odd degree?

4.3.2 Exercise. Show that if f and g are both invertible then $(g \circ f)^{-1} = g^{-1} \circ f^{-1}$.

4.3.3 Exercise. Using the previous exercise show that the following all have inverses:

1. $f(x) = \left(\frac{x-1}{x+1}\right)^5$.

2. $f(x) = \ln\left(\frac{x-1}{x+1}\right)$.

3. $f(x) = \frac{\ln(x)-1}{\ln(x)+1}$.

[Hint. First consider the basic building blocks appearing in the functions above, namely $f(x) = x^5$, $f(x) = \ln(x)$, and $f(x) = \frac{x-1}{x+1}$.]

4.4 The Natural Exponential Function

As we saw a couple of sections ago, the natural logarithm $\ln(x)$ is strictly increasing³. Hence, if you've done the relevant exercise, you already know that it has an inverse.

³ Well, we saw that its derivative is, by definition $\frac{1}{x}$, which on $(0, \infty)$ is always positive.

4.5 General Logarithms and Exponents

We have now *formally* defined what the function e^x is. From this, we want to define new functions a^x , for every $a \in \mathbb{R}_{>0}$. We want our general exponentiation functions to satisfy the following:

$$(a^b)^c = a^{bc},$$

for all $b, c \in \mathbb{R}$. This essentially forces our hand, because we already know that

$$(e^{\ln(a)})^r = a^r.$$

Thus our formal definition is:

$$a^x : \mathbb{R} \rightarrow \mathbb{R}$$

4.6 Hyperbolic Functions – EXTRA TOPIC

4.7 Inverse Trigonometric Functions

5

L'Hôpital's Rule

5.0.1 Theorem (L'Hôpital's Rule). Let $f, g : [a, b] \rightarrow \mathbb{R}$ be differentiable functions on (a, b) . Suppose that $g'(x) \neq 0$ for all $x \in (a, b)$. If:

- $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$

or

- $\lim_{x \rightarrow a} g(x) = \pm\infty$

Then:

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)},$$

provided $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \in \mathbb{R} \cup \{\pm\infty\}$.

5.0.2 Remark. The analogous statement holds for $\lim_{x \rightarrow b}$.

First, we give a proof assuming that $f(a) = g(a) = 0$ and both f' and g' are continuous at a .

Proof (simple case). We have that:

$$\begin{aligned} \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow a} \frac{f(x) - f(a)}{g(x) - g(a)} = \lim_{x \rightarrow a} \frac{\frac{f(x) - f(a)}{x - a}}{\frac{g(x) - g(a)}{x - a}} \\ &= \frac{\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}}{\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a}} = \frac{f'(a)}{g'(a)} = \frac{\lim_{x \rightarrow a} f'(x)}{\lim_{x \rightarrow a} g'(x)} \\ &= \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}. \end{aligned}$$

Note that in the first equality we used that $f(a) = g(a) = 0$, and in the equality

$$\frac{f'(a)}{g'(a)} = \frac{\lim_{x \rightarrow a} f'(x)}{\lim_{x \rightarrow a} g'(x)}$$

we used that $f'(x)$ and $g'(x)$ are continuous at a . □

5.0.3 Fact (Generalised Mean Value Theorem). Let $f, g : [a, b] \rightarrow \mathbb{R}$ be continuous functions, differentiable on (a, b) . Then, there is some $c \in (a, b)$ such that:

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}.$$

In particular,¹ if we take $g(x) = x$ we have that there is some point

¹ This is the usual **Mean Value Theorem**.

$c \in [a, b]$ such that:

$$f'(c) = \frac{f(b) - f(a)}{b - a}.$$

Now for the good and proper proof:

Proof. We will only give the proof in the case $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} g(x) = 0$. The argument when $\lim_{x \rightarrow a} g(x) = \pm\infty$ is similar, but requires a couple more calculations.

Without loss of generality,² we may assume that

² Since $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} = -\lim_{x \rightarrow a} -\frac{f'(x)}{g'(x)}$.

$$\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \neq +\infty.$$

Let $q, r \in \mathbb{R}$ such that $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} < r < q$. Since $\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} < r$, there is some point $c \in (a, b)$ such that for all $x \in (a, c)$ we have that:

$$\frac{f'(x)}{g'(x)} < r.$$

By the generalised Mean Value Theorem, for any $x, y \in (a, c)$ with $x < y$ there is a point $t \in (x, y)$ such that:

$$\frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(t)}{g'(t)} < r$$

Now taking $\lim_{x \rightarrow a}$ in the equation above we have that:

$$\frac{f(y)}{g(y)} \leq r < q.$$

s

□

Summary 2

6

Techniques of Integration

6.1 Basic Integrals

Before we delve deeper into more complicated integration techniques, it is important to have a list of the integrals we know:

- $\int x^r \, dx = \frac{1}{r+1} x^{r+1} + C$, for $r \neq -1$.
- $\int \frac{1}{x} \, dx = \ln |x| + C$

Summary 3

7

Sequences and Series

Summary 4

8

Curves on the Plane

9

Conic Sections

Summary 5

Index

- antiderivative, 7
- codomain, 36
- Darboux sum, 7
- derivative, 5
 - n -th, 7
 - second, 7
- differentiable, 5
- domain, 36
- energy
 - kinetic, 24
 - potential, 25
- function, 36
 - bijective, 36
 - injective, 36
 - inverse, 36
 - strictly decreasing, 37
 - strictly increasing, 37
 - strictly monotone, 37
 - surjective, 36
- integral
 - definite, 8
 - indefinite, 10
- length
 - of a curve, 20
- logarithm
 - natural, 35
- Mean Value Theorem, 41
- moment, 25
 - of a function, 26
- normal line, 5
- parametrised curve, 29
- partition
 - of an interval, 7
- range, 36
- Riemann sum
 - left, 7
 - right, 7
- Riemann sum corresponding to the partition P , 7
- shell method, 19
- solid of revolution, 16
- tangent line, 5
- work, 23